



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals & Interfacing

Batch: 18th

Semester: Fall 2022

Date: 18/03/2023

Final Examination

Time: 2 hour

Total Marks: 40

Answer any five questions from the followings:

5x8= 40

- 1 (a) What are the main differences between a hard disk drive (HDD) and a solid-state drive (SSD) in terms of their technology, speed, durability, and cost? 4
(b) Draw the block diagram and explain the operating modes of PPI 8255. 4
- 2 (a) What are the working principles behind a gas discharge display and how does it generate light? 4
(b) What are the key components of a NIC and how do they work together to enable communication between a computer and a network? 4
- 3 (a) What are the key differences between LED and LCD displays, and how do they impact their performance and functionality? 4
(b) How do the pins of HT82K68E correspond to its internal components and functions? 4
- 4 (b) What are the different operating modes of the PPI 8255, and how do they differ? 4
(c) How does DMA controller 8257/8237 enable DMA operations in a computer system? 4
- 5 (a) Can a data converter be classified as an input or output device? 2
(b) What are the limitations and drawbacks of using a non-programmable port, and how do they compare to those of a programmable port? 3
(c) How does latching work in the context of the 8212 latch? 3
- 6 (a) Why is a PPI needed in a computer system or electronic device? 2
(b) Draw the block diagram of 8155 with 8055. 3
(c) Write short note on the following terms: 2x1.5 =3
 - i. Bidirectional buffer
 - ii. Control Buffer
- 7 (a) What types of data can be displayed using 7, 9, 14 and 16 segments? 2
(b) How does the implementation of zone-based storage allocation in hard disks impact the read/write performance, storage capacity utilization, and reliability of the drive? 3
(c) What are the potential trade-offs and benefits associated with zoning in modern hard disk technology? 3



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 325: System Analysis & Design

Batch: 18th

Date: 22/03/2023

Final Examination

Semester: Fall 2022

Time: 2 hours

Total Marks: 40

Part : B

Time: 1 hour 50 Minutes

Total Marks: 30

(Answer any 05 (Five) questions from the rest)

[The figures in the right margin indicate full marks for the respective question]

1. a) Define system requirements. *ek-4(5)* [2]
b) Show the analysis phase activities and key questions. *4(4)* [4]
2. a) How will we assure the quality assurance in SAD? *ca* [3]
b) Draw the summary types of requirements. *- ca* [3]
3. a) Mention the characteristics of prototypes. [2]
b) Show some models created during design phase. *o* [4]
4. a) Draw the diagram of data entities compared to objects. *-* [3]
b) Explain the context diagram with the example of course registration system. *—* [3]
5. a) What are the important components of the environment that will affect the project? *-* [3]
b) How will we generate alternatives for implementation? *Show it by figure. —* [3]
6. a) What are the steps for creating the structure chart from the DFD? *-ca* [3]
b) How will you show the system sequence diagram for create new order? *-c10* [3]



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 323: Operating System

Batch: 18th

Semester: Fall 2022

Date: 27/01/2023

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions of the followings.

5 × 8 = 40

1. (a) Explain the reasons for which processes need to communicate with other process. What are the mechanisms to establish communication between processes? 4
- (b) Write the strict alternation algorithm of mutual exclusion. Explain the disadvantages this algorithm has. 4
2. (a) Why do we need higher-level synchronization to remove race condition? What are the most common mechanisms to ensure higher-level synchronization? 3
- (b) Draw a Gantt chart to show the execution of the following processes inside the critical region while all the processes are synchronized using counting semaphore and the initial value of the semaphore is 2: 5

Process	Arrival Time	Burst Time
P ₁	10	15
P ₂	5	10
P ₃	0	20
P ₄	12	22
P ₅	19	17
P ₆	25	13
P ₇	30	15

3. (a) By using the following scheduling algorithm, draw the CPU scheduling Gantt chart: 5

Process	Burst Time	Arrival Time	Algorithm
P ₁	40	0	RR interval: 6
P ₂	25	20	
P ₃	35	15	
P ₄	40	0	SRTF
P ₅	10	10	
P ₆	30	60	
P ₇	20	90	

- (b) Complete the table given below for the above-mentioned Gantt chart. 3

Waiting time	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇
Turnaround time	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇

4. (a) When CPU scheduler does need to take its decision? Explain them briefly. 4
 (b) Describe First Come First Serve Scheduling algorithm. Explain its limitations. 4
5. (a) List the mechanisms to allocate data in secondary memory. Explain them with their limitations. 4
 (b) What is the total size in KB of index Blocks that are used to store 7GB of data in Hard Disk using Indexed allocation mechanism? 4
6. (a) Why do we need to predict the burst time of a process? Derive the equation to predict the Burst time of the next process. Generalized the equation for $\alpha = 0$ and $\alpha = 1$. 4
 (b) Explain that Shortest job first is a priority scheduling algorithm. What is the problem of priority scheduling? State its solution. 4
7. (a) Suppose we want to access a data from the hard-disk. To access the data the head pointer needs to move 4, 24, 77, 107, 119, 15, 65, 75, 100, 180, 64, 155, 88, 136, 75, 67, 99 number of tracks. Disk tracks starts from 0 to 199. Initially the head is in 25th number track. Find the total cylinder/track number, the head has to move in SCAN and C-LOOK Disk scheduling algorithm. 4
 (b) Consider the above question to calculate the minimum access time to transfer 17KB of data for a disk with 1KB/sector. The read/write arm needs 0.001ms to pass one cylinder/track on the disk. The disk rotates at 6400RPM and transfer data at a rate of 6MB/s. The controller overhead is 2ms and there are 3 processes in the device queue each need 2s to complete their tasks. 4



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323

Course Title: Numerical Analysis

Final Examination

Semester: Fall 2022

Date: 3/25/2023

Total Time: 2 Hours.

Total Marks: 40

04 - APR - 2023

Answer any four (04) of the following questions:

4 × 10 = 40

1. (a) Derive the Euler method for first order differential equation. [4]

(b) Find the approximate solution of $\frac{dy}{dx} = x^2 + y^2$; $y(0) = 0$ for $x = 0.5$ by using Euler method. [6]

Page 3 → 5

2. (a) Derive the second order Runge-Kutta method for first order differential equation. [4]

(b) Find the approximate solution of $\frac{dy}{dx} = \frac{x-y}{x+y}$; $y(0) = 1$ for $x = 2$ by using 2nd order Runge-Kutta method. [6]

Page 17 → 20

3. (a) Derive the Newton-Raphson method for non-linear system of differential equation. [5]

(b) Solve the following system of non-linear equation by using fixed-point method [5]

$$x^2 + y = 11.$$

$$y^2 + x = 7.$$

Page - 29 → 31

4. (a) Write the geometrical representation of trapezoidal rule. [3]

(b) Find the approximate solution of $I = \int_4^{5.2} \ln(x) dx$ by using Weddle's method. [7]

5. Find the approximate solution of $I = \int_0^1 \frac{1}{1+x^2} dx$ by using (Trapezoidal rule, Simpson's-
value $\frac{1}{3}$ rule, Simpson's- $\frac{3}{8}$ rule.) Also find the error of the above method with exact value of

$$I = \int_0^1 \frac{1}{1+x^2} dx = \frac{\pi}{4}.$$

6. (a) Write the application of numerical analysis in real life. [2]

(b) Find the approximate solution of $\frac{d^2y}{dx^2} = x \left(\frac{dy}{dx}\right)^2 - y^2$; $y(0) = 1$, $\frac{dy(0)}{dx} = 0$ for $x = 0.3$ [8]
by using 4th order Runge-Kutta method.



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 321: Theory of Computing

Batch: 18th

03/04/2023
Date: 15/03/2023

Final Examination

Semester: Fall 2022

Time: 2 Hours

Total Marks: 40

Answer any five questions of the followings.

5 x 8 = 40

1. (a) Why Context-free grammar (CFG) is necessary in Automata? 2
(b) Consider the following context-free Grammar G: 6x1=6

$$\begin{aligned}R &\rightarrow XRX \mid S \\S &\rightarrow aTb \mid bTa \\T &\rightarrow XTX \mid X \mid \epsilon \\X &\rightarrow a \mid b\end{aligned}$$

Now answer all the following questions:

- What are the variables of G?
 - What are the terminals of G?
 - Which is the start variable of G?
 - Give three strings in L(G).
 - Give three strings not in L(G).
 - Give a description of L(G).
2. (a) Convert the following context-free grammar into an equivalent grammar in Chomsky normal form. Show the steps of your conversion. Here, 0 and 1 are the terminals. 4

$$\begin{aligned}S &\rightarrow AIA \mid B \\A &\rightarrow 0A \mid 1A \mid \epsilon \\D &\rightarrow 00\end{aligned}$$

- (b) Determine whether 01101 is in the language of the 2(a) grammar thus found using the CYK algorithm. 4

3. (a) Consider the following grammar G: 4
 $S \rightarrow 0A \mid 1B$
 $A \rightarrow 0AA \mid 1S \mid 1$
 $B \rightarrow 1BB \mid 0S \mid 0$

Is the given grammar ambiguous or unambiguous? Explain.

- (b) Write the context free grammar for the following language: 4
"All binary strings with both an even number of zeroes and an even number of ones."

4. (a) Write the formal definition of Push Down Automata (PDA). Briefly explain the components of a PDA. 3

- (b) Design a Push Down Automata (PDA) that recognizes the language

5

$$L = \{a^n b^n c^m d^m \mid n \geq 1, m \geq 1\}$$

Draw its transition diagram and write transition functions.

- 5 (a) Consider the following grammar (G):

2

$$S \rightarrow TU$$

$$U \rightarrow 11$$

$$T \rightarrow TV \mid \varepsilon$$

$$V \rightarrow 00 \mid 11$$

Write the regular expression for the grammar (G).

- (b) Consider the following production rules for the grammar (H):

6

$$S \rightarrow xB \mid yA$$

$$A \rightarrow xS \mid yAA \mid x$$

$$B \rightarrow yS \mid xBB \mid y$$

Write the left-most derivation and parse tree for the input string 'xxxxyxyxyx'.

6. (a) Write the formal definition of Turing Machine. Draw the relational diagram among finite automata, push down automata and Turing machine.

2

- (b) Consider the grammar consisting of 7 productions

6

$$S \rightarrow aA \mid aBB$$

$$A \rightarrow aaA \mid \varepsilon$$

$$B \rightarrow bB \mid bbC$$

$$C \rightarrow B$$

Simplify the given grammar using elimination of unit, useless and null productions.

7. (a) If the input string size of a Moore machine is 'n' then what is the size of the output string? Design a Moore machine to find the 1's complement of a given binary number.

4

- (b) Design a Mealy machine for a binary input sequence such that if it has a substring 101, the machine output A, if the input has substring 110, it outputs B otherwise it outputs C.

4